PES UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ACADEMIC YEAR 2025-2026 | FIFTH SEMESTER B.TECH

SYSTEM VALIDATION REPORT

(System, Non-Functional and Security Testing)

PROJECT: SPRINTFLOW – PROJECT MANAGEMENT SYSTEM

(Workflow, Visibility, Estimation, and Kanban)

SUBMITTED BY: TEAM 11 (SECTION 5C)

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Version: 1.1 (Final Submission) | Date: 11-11-2026 | Testing Phase: 02-11-2026 to 06-11-2026

Repository: https://github.com/Darshanpawar7/SE-SRS

Document: System Validation Report, Testing Phase (Part 3, item 2.b.ii)

Report owner: Dhatri Shivaprasad, Test Engineer (Frontend) (GitHub issue #92)

Prepared by: Team 11 (Section 5C, PES University)

Date: 11-11-2026

Build under test: `main` branch, commit `f9e0727`; defect retest on PR #98, commit `e8f5905`; final regression on `main` at commit `2b8fdbc`; pre-submission regression on `main` at commit `970c656` (11-11-2026, Section 12.3)

Commit IDs: On 11-11-2026 some commit messages in the repository history were edited. The code did not change, but the commit IDs did. This report keeps the IDs recorded during testing; on GitHub today `f9e0727` is `2398c89`, `e8f5905` is `1f84804` and `2b8fdbc` is `21f3681`. GitHub Actions runs from before that date still show the old IDs.

Approved by: Darshan P Pawar, Test Architect (STP Section 9), 11-11-2026

Repository location: `docs/test-reports/SYSTEM_VALIDATION_REPORT.md`, test script `tests/system/system-validation.mjs`, raw results in `docs/test-reports/evidence/`

Revision History

Version	Date	Author	Change Summary
1.0	06-11-2026	Team 11	System test results for every case in the Test Case Specification, the STP security checks, cross-browser results, two defects with their retests, and the requirements traceability
1.1	11-11-2026	Team 11	Added the pre-submission regression on the final `main` branch (GitHub Actions run 42973669659): the browser suite in Chrome, Edge and Firefox, the four cases that were blocked in the testing phase, run on a disposable Supabase database (Section 6.1), and a Lighthouse accessibility measurement for PMS-NF-003 (Section 12.3). Added the entry criteria (Section 2.1) and the defect density, defect aging and requirement coverage metrics (Section 3). Corrected details found in a final review: the XSS result wording, the views opened in TC-AVAIL-01, the scope of SEC-TOKEN and SEC-FUZZ, and the Test Summary Report status. Updated the verdict, the traceability, the exit criteria and the conclusion, and noted the changed commit IDs. The results of the testing phase are unchanged.

1. Introduction

1.1 Purpose

This report checks the complete SprintFlow system against its requirements. Every case in the Test Case Specification was run end to end in real browsers against the production build, along with the security checks listed in STP Section 5.4. It covers the system test level of the Software Test Plan (Section 5.1) and the non-functional cases in STP Section 13.2.

1.2 Scope

In scope are the 26 cases in the Test Case Specification (20 functional and 6 non-functional), two supplementary cases we added for SRS requirements that had no matching case (PMS-F-002, and PMS-F-007, whose SRS traceability entry points to TC-TASK-02, a priority case), and five security checks. Unit and integration results are in the Unit Test Report; code and branch coverage are in the Code Coverage and Branch Coverage Report.

1.3 References

- Software Test Plan, STP v1.0 (`docs/STP_Document_Team11.md`), Sections 5, 11, 13 and 14
- Test Case Specification (`docs/TEST_CASE_SPECIFICATION.md`)
- Software Requirements Specification (`docs/SRS_Document_Team11.md`)

- Sprint 1 deviations and ID mapping (`docs/SPRINT1_DEVIATIONS.md`)
- Earlier defects DEF-01 to DEF-03 (`docs/TEST_EXECUTION_AND_CICD.md`)

2. Test Environment and Approach

Item	Value
Build under test	`main` at commit `f9e0727`, built with `npm run build` and served with `vite preview`
Data mode	Zero-configuration (localStorage) mode with the seed data in `src/services/mockData.js`. No Supabase credentials were supplied to the browser suite, so none of its tests wrote to a real database.
Browsers	Google Chrome 157.0.8103.38 (primary), Microsoft Edge 156.0.4314.58, Firefox 142.0.1 (Playwright build)
Screen size	1440 x 900
Machine	Windows 11 Pro laptop with Node.js 24.15.0 (main run, retest and final regression of 06-11-2026)
Test driver	Playwright 1.56.1, script `tests/system/system-validation.mjs` (`npm run test:system`)
Main run	06-11-2026, 17:37 IST to 17:39 IST, one uninterrupted run
Defect retest	06-11-2026, 21:37 IST, build of PR #98 (commit `e8f5905`), Chrome
Final regression	06-11-2026, 23:05 IST, final `main` (commit `2b8fdbc`), Chrome, Edge and Firefox
Pre-submission regression	11-11-2026, 12:08 IST, final `main` (commit `970c656`), GitHub Actions run 42973669659 on Ubuntu 24.04: Chrome, Edge and Firefox, plus a disposable Supabase database for the cloud cases (Sections 6.1 and 12.3)
Raw results	`docs/test-reports/evidence/system-validation-results.json`

The specification marks most cases as manual. For this phase we wrote each manual step as a browser action in the Playwright script, and each expected result as a check that compares it with what the page actually shows. This made the run repeatable, and the actual result of every expected result is recorded rather than ticked by hand. The rules we followed:

- Every case starts in a fresh browser context: empty storage, the seed data, signed in as the Team Lead (Darshan P Pawar). The order of the cases cannot change the results.
- A case passes only if every one of its checks passes, as Section 2 of the specification requires.
- A case whose precondition cannot be met is recorded as Blocked, together with any steps that could still be run.
- A screenshot is saved for every failure.

2.1 Entry criteria (STP Section 5.3)

Entry criterion	Status	Evidence
Stable build delivered (`npm run build` completes without errors)	Met	The production build of `f9e0727` succeeded (143 modules), and every case ran against it through `vite preview`.
All unit tests passing (`npm test` reports 0 failures)	Met	234 of 234 unit tests passed on the same commit (Unit Test Report, Section 4).
Test environment ready (Node.js 18+, browser available)	Met	Node.js 24.15.0 with Chrome 157, Edge 156 and Firefox 142 (table above).
Test data available (seed data in `mockData.js`)	Met	Every case starts from the seed data in a fresh browser context.
SRS and SAD documents reviewed and approved	Met	The SRS is marked "Approved for Academic Submission" and the SAD "Final Submission / Approved".

3. Execution Summary

Group	Cases	Pass	Fail	Blocked
Test Case Specification, functional (TC-AUTH-01 to TC-LOG-01)	20	17	1	2
Test Case Specification, non-functional (TC-PERF-01 to TC-RECOV-01)	6	5	0	1
Supplementary cases for SRS requirements with no specified case	2	2	0	0
Security validation, STP Section 5.4	5	4	0	1
Total	**33**	**28**	**1**	**4**

Metric (STP Section 14.1)	Result	Target
Cases executed in full (pass + fail)	29 of 33 (87.9%)	100%
Cases with at least one step executed	31 of 33 (93.9%)	–
Pass rate of executed cases, main run	28 of 29 (96.6%)	95% or higher
Pass rate after the DEF-04 fix was retested (Chrome only, Section 5.2)	29 of 29 (100%)	95% or higher
Pass rate in the final regression on `main` (Section 12)	29 of 29 (100%)	95% or higher
Cases executed in full, pre-submission regression on 11-11-2026 (Sections 6.1 and 12.3)	33 of 33 (100%)	100%
Pass rate, pre-submission regression on 11-11-2026	33 of 33 (100%)	95% or higher
Open critical or high-severity defects	0	0
Defect density	2 defects in 2 modules (Kanban board, workload panel): at most 1 per module	Less than 2 per module
Defect aging (discovery to resolution)	DEF-04: 5 h 13 min; DEF-05: 1 h 01 min; average 3 h 07 min	Less than 24 hours
Requirement coverage	28 of 28 SRS requirements have at least one test case or objective evidence (Section 8)	100%

Defect aging runs from the opening of each defect's GitHub issue to its closure by the merged fix: #95 from 17:30 to 22:43 IST and #100 from 21:54 to 22:55 IST, both on 06-11-2026.

Verdict: Pass. Both defects found were fixed and retested, and the four cases that were blocked in the testing phase passed on 11-11-2026. Every case that could run without a cloud database passed in the main run, apart from one usability detail in filtering (DEF-04, Section 5). Dhatri fixed DEF-04 in PR #98, and a wording slip found in the workload panel (DEF-05) was fixed in PR #101; both retests passed. A full regression run on the final build, in all three browsers, passed with no failures (Section 12). The cases that needed a Supabase project (Section 6) were run on 11-11-2026 against a disposable Supabase database and passed, so all 33 cases now pass (Sections 6.1 and 12.3).

4. Results by Test Case

Requirement IDs follow the SRS, our baseline. The Test Plan ID for the same feature is shown next to it (mapping in `docs/SPRINT1_DEVIATIONS.md`, Section 1). The full expected-versus-actual result of every check is in Appendix A.

#	Test case	SRS ID	Test Plan ID	Title	Result	Key evidence (actual result)
1	TC-AUTH-01	PMS-F-001	PMS-F-001	Authenticate and establish session	**Pass**	Top bar shows Darshan / Team Lead; still signed in after refresh

#	Test case	SRS ID	Test Plan ID	Title	Result	Key evidence (actual result)
2	TC-PROJ-01	PMS-F-002	–	Show project metadata (supplementary)	**Pass**	Key PMS, project name and active sprint in the top bar; project dates 2026-09-01 → 2026-10-30
3	TC-TASK-01	PMS-F-006	PMS-F-002	Create a task with complete metadata	**Pass**	Card shows title, High badge, 5 SP and Dhatri; with Backlog chosen it tops Backlog; survives refresh (default column: Section 7)
4	TC-TASK-02	PMS-F-006	PMS-F-016	Assign Critical priority	**Pass**	Badge reads "Critical" as text
5	TC-TASK-03	PMS-F-006	PMS-F-002	Reject a task with an empty title	**Pass**	Empty: browser validation: "Please fill out this field."; one space: inline message: "Enter a task title and choose a valid priority."; 6 cards before and after
6	TC-LABEL-01	PMS-F-007	–	Add labels to a task (supplementary)	**Pass**	Pills #feature #ui #api; unchanged after refresh
7	TC-KAN-01	PMS-F-003	PMS-F-003	Render the five-column board	**Pass**	Backlog, To Do, In Progress, Code Review, Done; every count badge matches its cards
8	TC-KAN-02	PMS-F-004	PMS-F-004	Transition a task by drag-and-drop	**Pass**	Card in In Progress; Backlog 1 → 0, In Progress 2 → 3; kept after refresh
9	TC-KAN-03	PMS-F-005	PMS-F-005	Advance a task by single action	**Pass**	To Do → In Progress; 0 enabled advance controls in Done
10	TC-KAN-04	PMS-F-004	PMS-F-004	Cancel a drag outside any column	**Pass**	1 dragstart, 0 drop events; card stayed in In Progress
11	TC-FILT-01	PMS-F-017	PMS-F-011	Filter by priority	**Fail**	An explicit empty state names the active filters: five columns each show "Drop tasks here"; no message names the active filters
12	TC-EST-01	PMS-F-008	PMS-F-006	Assign Fibonacci story points	**Pass**	Choices 1, 2, 3, 5, 8, 13, 21; each value saved; capacity 33 → 49 (task 5 → 21)
13	TC-EST-02	PMS-F-009	PMS-F-008	Calculate sprint capacity	**Pass**	Capacity 16 with 3+5+8, then 11 after removing the 5
14	TC-SPR-01	PMS-F-010	PMS-F-007	Create a sprint	**Pass**	Sprint 3 listed with goal and range 2026-11-06 → 2026-11-19; offered in the task dialog
15	TC-BURN-01	PMS-F-011	PMS-F-009	Update the burndown chart	**Pass**	Today's actual 20 → 15 (task 5 SP); ideal line unchanged
16	TC-DASH-01	PMS-F-012	PMS-F-010	Display KPI metrics	**Pass**	total 33, delivered 13, completion 39%, remaining 20; matches the board
17	TC-WORK-01	PMS-F-013	PMS-F-017	Show workload distribution	**Pass**	10 + 8 + 5 + 10 = 33; sprint total 33
18	TC-DOCS-01	not in SRS	PMS-F-018	Render in-app documentation	**Pass**	All three tabs render (3172 characters, 1093 characters, 947 characters)
19	TC-SUPA-01	PMS-F-015	PMS-F-013	Persist to Supabase PostgreSQL	**Blocked**	Precondition not met: needs a Supabase test project with database/schema.sql (and the 20261103 migration) applied.
20	TC-OFF-01	PMS-F-016	PMS-F-014	Persist offline without configuration	**Pass**	2 of 2 present after reopening the tab; top bar shows Local / Demo Mode
21	TC-SET-01	PMS-F-015	PMS-F-015	Switch data mode from settings	**Blocked**	Step 3 passed: malformed URL rejected, offline mode kept. Steps 1–2 need a Supabase test project (Section 6)
22	TC-LOG-01	PMS-F-014	PMS-F-012	Record an audit trail for mutations	**Pass**	Records 4 → 7: CREATE, UPDATE, DELETE, each with user, task and timestamp
23	TC-PERF-01	PMS-NF-001	PMS-NF-001	Drag-and-drop transition under 50 ms	**Pass**	10 measured drops after 2 warm-ups: p50 2.3 ms, p95 6.6 ms, max 6.6 ms
24	TC-AVAIL-01	PMS-NF-002	PMS-NF-002	Stay usable with the network disabled	**Pass**	With the network off: the Overview, Sprints, Work items and Board views open, a task is created, top bar shows Local / Demo Mode
25	TC-COMPAT-01	not in SRS	PMS-NF-003	Core workflows on Chrome, Edge and Firefox	**Pass**	Same results and 0 px layout drift; chrome 157.0.8103.38, edge 156.0.4314.58, firefox 142.0.1

#	Test case	SRS ID	Test Plan ID	Title	Result	Key evidence (actual result)
26	TC-SEC-01	PMS-SR-003	PMS-NF-004	Encrypted transport to Supabase	**Blocked**	An http:// project URL is refused. Live HTTPS traffic not inspected: needs Supabase (Section 6)
27	TC-LOAD-01	not in SRS	PMS-NF-005	Board with 100+ tasks	**Pass**	Board ready 162 ms after navigation with 106 tasks; drop-to-render p95 8.9 ms over 5 drops
28	TC-RECOV-01	PMS-NF-004	PMS-NF-006	No data loss across disconnect and reconnect	**Pass**	Tasks created offline still present after reconnect and reload (2 of 2 present)
29	SEC-XSS	PMS-SR-004	STP 5.4	XSS input sanitisation	**Pass**	Script tags and event handlers in title, description and sprint goal never ran; the tags were removed and only the text kept
30	SEC-RLS	PMS-SR-001	STP 5.4	Row Level Security on direct API calls	**Blocked**	Needs a live Supabase project to send cross-project API calls. The policies in database/schema.sql are checked statically by src/tests/databaseSecurity.test.js.
31	SEC-TOKEN	not in SRS	STP 5.4	Session token handling	**Pass**	Neither the password nor any JWT is kept in browser storage (expired tokens: SEC-EXPIRED, Section 12.3)
32	SEC-KEYS	PMS-SR-002	STP 5.4	API key exposure in the client bundle	**Pass**	No complete JWT or Supabase secret key in the bundle (0 found in 2 files)
33	SEC-FUZZ	PMS-SR-004	STP 5.4	Input validation fuzzing	**Pass**	Title field: 5,000 characters bounded (120 stored), no crash; special characters shown as typed

5. Defects Found

Defect IDs continue the series in `docs/TEST_EXECUTION_AND_CICD.md` (DEF-01 to DEF-03).

ID	Found by	Severity	Summary	Issue / fix	Status
DEF-04	TC-FILT-01, step 2 (first seen in trial runs of the script, confirmed in the main run)	Low (usability)	When the board's filters match no task, the five columns show "Drop tasks here" and nothing says that filters are hiding the work, or which filters are active. "Reset Filters" is shown and restores the board.	Issue #95; fixed by Dhatri Shivaprasad in PR #98	Fixed and retested; PR #98 merged on 06-11-2026 and issue #95 closed
DEF-05	Review of TC-WORK-01 evidence	Cosmetic	The Capacity & Workload panel reads "1 tasks" for a member with one task, because `WorkloadList.jsx` always prints the plural. TC-WORK-01 still passes; the totals are correct.	Issue #100; fixed in PR #101	Fixed and retested (Section 12.1); issue #100 closed

5.1 DEF-04 in detail

Expected (TC-FILT-01): "an explicit empty state names the active filters and offers to clear them; columns are not left silently blank."

Actual (main run): the High priority filter itself works; step 1 passed with 3 High tasks shown. After a search term that matches nothing was added in step 2, the board looked exactly like an empty sprint (Figure 1). The Work items view already shows "No work items match." in the same situation, so the board was the only place affected. No data was lost or changed.

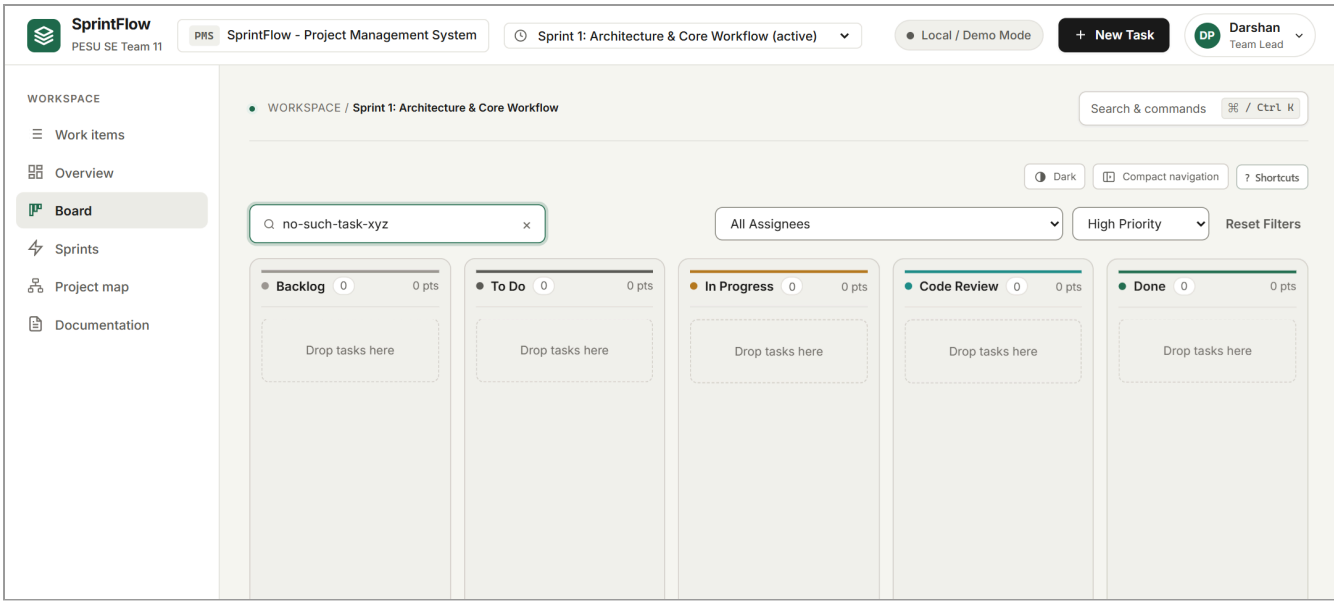


Figure 1: DEF-04 on the tested build (f9e0727): no message, five empty columns.

5.2 Retest of the fix (PR #98)

PR #98 adds a status message above the board that names the active filters and offers a Reset Filters button. It also adds three unit tests and extends the existing reset test. We built the PR (commit `e8f5905`) and ran TC-FILT-01 again in Chrome on 06-11-2026. All four checks passed:

Expected result checked	Actual result	Verdict
Only High priority tasks are visible	3 visible: high, high, high	**Pass**
No task matches the combined filters	0 cards visible	**Pass**
An explicit empty state names the active filters	Message shown: "No tasks match the active filters: Priority: High; Search: “no-such-task-xyz”." (Figure 2)	**Pass**
The empty state offers to clear the filters	"Reset Filters" button shown	**Pass**

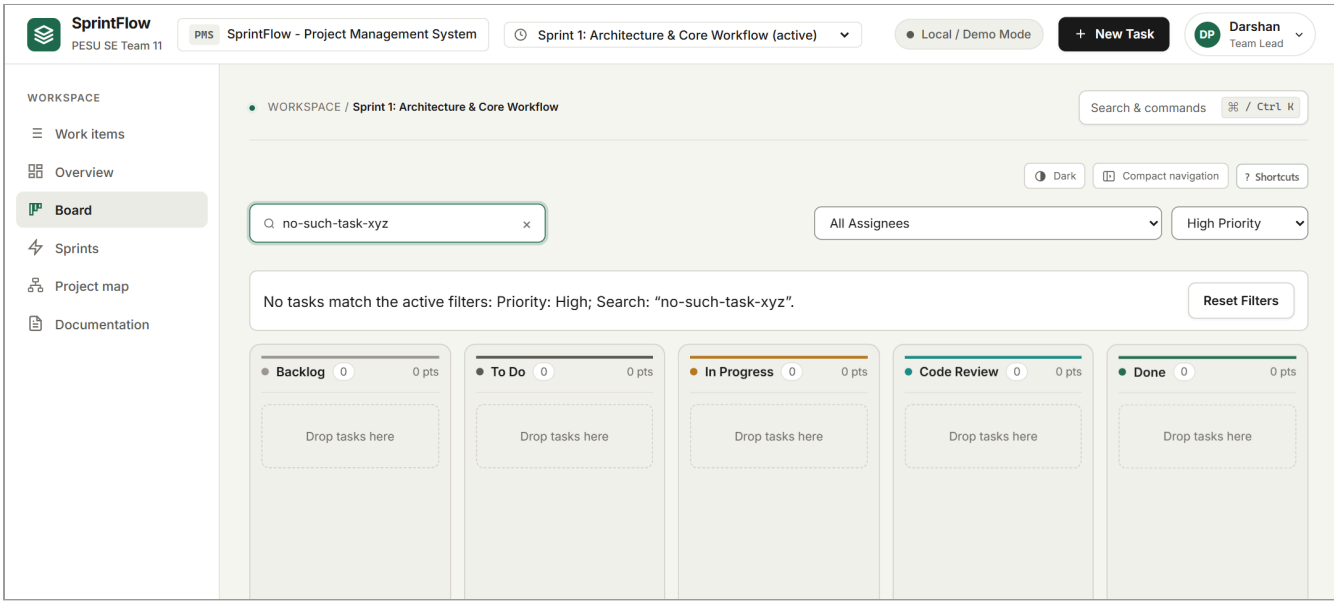


Figure 2: Retest on the fix build (PR #98): the empty state names the filters and offers Reset Filters.

To make sure the fix broke nothing else, we ran the whole Chrome suite on the same build: 29 passed, 0 failed and 4 blocked, the same four cloud cases as in the main run. On the fix branch, `npm test` passed 237 of 237 tests (the 234

existing tests plus the new ones), and lint and the production build both succeeded. In this Chrome-only run TC-COMPAT-01 could only compare Chrome with itself; the three-browser comparison was repeated in the final regression (Section 12).

PR #98 also rewrote TC-FILT-01 in the Test Case Specification as a four-step case, "Combine filters and report an empty result". The retest above covers its step 2 in the browser and confirms that the Reset Filters action is offered. Clearing the filters with that action (step 3) was seen working in the main run, and steps 1 and 4, where no message may appear, are checked by the new unit tests in `kanbanEmptyState.test.jsx`.

5.3 DEF-05

While checking the screenshots for this report we noticed that the workload panel says "1 tasks • 8 story points assigned" for Dhatri and "1 tasks • 5 story points assigned" for Chandan (workload screenshot in Appendix B). The text is built as `{taskCount} tasks` on lines 20 and 41 of `WorkloadList.jsx`. It does not affect any number, so TC-WORK-01 passes, but it is visible to every user. It was logged as issue #100 and fixed in PR #101 (retest in Section 12.1).

No other defects were found. No uncaught JavaScript error occurred in any case, the XSS and fuzzing inputs neither ran nor crashed the page, and every change survived a page refresh.

6. Cases Blocked in the Testing Phase: Supabase Project Needed

These cases have a live Supabase project as their precondition (the STP lists this as a dependency in Section 11). We did not run them against the team's live database, because their steps write rows to it. The table shows what could be checked without one during the testing phase; the rest was tracked in issue #96 and run later on a separate, disposable Supabase database (Section 6.1).

Case	SRS	Verified now	Still to run on a Supabase test project	Supporting automated evidence
TC-SUPA-01	PMS-F-015	–	Create a task while connected, confirm the row in `tasks`, refresh and confirm it remains	`supabaseSync.test.js` (15 tests against a mocked client: reads, first-connection publish, column mapping, deletes, audit rows, failure handling)
TC-SET-01	PMS-F-015	Step 3: a malformed URL is rejected with "Enter a valid https:// Supabase project URL…", and the offline configuration is kept	Steps 1–2: valid credentials switch the indicator to cloud mode	`configSecurity.test.js`, `supabaseSync.test.js` (TC-SET-01 malformed-URL test)
TC-SEC-01	PMS-SR-003	A plain `http://` project URL is refused, so the client can only be configured for HTTPS	Inspect live requests: every request is HTTPS and no key appears in a URL	`configSecurity.test.js` ("accepts only HTTPS URLs…")
SEC-RLS	PMS-SR-001	–	From a second account, query another project's tasks through the REST API and expect zero rows	`databaseSecurity.test.js` ("enables RLS on every application table…")

Before these are run, `database/schema.sql` and then `database/migrations/20261103_work_signals.sql` must be applied to the test project. The migration has been needed since PR #87; without it every cloud task save is rejected.

6.1 Results on a disposable Supabase database (11-11-2026)

All four cases were run on 11-11-2026 in the pre-submission regression (GitHub Actions run 42973669659, Section 12.3). The workflow started a Supabase database and Auth service on the GitHub Actions runner with the Supabase CLI, applied `database/schema.sql` and the migration above, ran the cases against the production build in Chrome, and then deleted the database. Because this was not the team's hosted project, the hosted project's own certificate and settings were not part of the check.

Case	Date	Result	Observed
TC-SUPA-01	11-11-2026	**Pass**	A task created through the app was read back directly from PostgreSQL, and loaded again after the app's browser caches were cleared.
TC-SET-01 (steps 1–2)	11-11-2026	**Pass**	A valid https:// configuration was saved, the status indicator changed and a real Supabase Auth sign-in succeeded. A malformed replacement was refused and the working configuration was kept.
TC-SEC-01	11-11-2026	**Pass**	Auth and REST requests went over HTTPS with TLS 1.3, and no password, API key or access token appeared in any recorded request URL. The test used a one-day self-signed certificate on the runner.
SEC-RLS	11-11-2026	**Pass**	With two real accounts, reading rows of a project the account does not belong to returned zero rows, an insert was refused, updates and deletes changed nothing, and the private row stayed unchanged. A read without signing in was refused.

7. Differences Between the Specification and the Built System

These are not defects. Each one is a documented design change or a detail the specification leaves open. They will be carried into the update of the Test Case Specification (Part 3, item 1, "Test Cases Updation").

#	Case	Specification says	System does	How it was tested
1	TC-TASK-01	New card appears at the top of Backlog	A new task starts in **To Do** unless another status is chosen (documented in `SPRINT1_DEVIATIONS.md`)	Created with the default (card lands in To Do) and again with Backlog chosen (card tops Backlog). All other expected results checked on both.
2	TC-AUTH-01	Select the Team Lead profile and confirm	Sign-in is a profile plus its password (documented in `SPRINT1_DEVIATIONS.md`)	Signed in as Darshan P Pawar (Team Lead) with the demo password
3	TC-TASK-03	Both attempts rejected with an inline validation message	Empty title: the browser's required-field message ("Please fill out this field."). Single space: the app's inline message.	Both rejected and no task created, so the case passes. The wording should name both mechanisms.
4	TC-BURN-01	"Note the actual burn line"	The actual line is recorded once a day per browser, so a fresh browser has a single point	Read today's value from the chart's "Daily burndown values" table, before and after completing a 5-point task
5	TC-EST-02	"Remove the 5-point task"	–	Removed by deleting the task
6	TC-PERF-01	Measure with the Performance API	–	`performance.mark` at the drop event, measured to the DOM update that shows the card in its new column; 10 samples after 2 warm-ups
7	TC-AVAIL-01, TC-RECOV-01	Disconnect the network	–	Network disabled in the browser context, in zero-configuration mode. The cloud retry queue across a reconnect is covered by `persistenceRecovery` unit tests and, on 11-11-2026, by CLOUD-RECOVERY on a real database (Section 12.3).
8	TC-COMPAT-01	Chrome, Firefox and Edge	–	Desktop Firefox is not installed on the test machine, so Playwright's Firefox 142 build was used
9	Section 10 of the specification	Lists 2 automated test files	35 unit test files (234 tests) plus this system suite	–
10	Section 11 of the specification	–	SRS PMS-F-002 (project metadata) and PMS-F-007 (labels) have no system case	Supplementary cases TC-PROJ-01 and TC-LABEL-01 were added; both pass

8. Requirements Traceability (SRS Baseline)

SRS ID	Requirement	Verified by	Result
PMS-F-001	Sign-in and profile switching	TC-AUTH-01	Pass
PMS-F-002	Project metadata	TC-PROJ-01 (supplementary)	Pass

SRS ID	Requirement	Verified by	Result
PMS-F-003	Five Kanban columns	TC-KAN-01	Pass
PMS-F-004	Drag and drop	TC-KAN-02, TC-KAN-04	Pass
PMS-F-005	One-click stage advance	TC-KAN-03	Pass
PMS-F-006	Create and update tasks	TC-TASK-01, TC-TASK-02, TC-TASK-03	Pass
PMS-F-007	Labels and tags	TC-LABEL-01 (supplementary)	Pass
PMS-F-008	Fibonacci story points	TC-EST-01	Pass
PMS-F-009	Planned points and completion	TC-EST-02, TC-DASH-01	Pass
PMS-F-010	Sprint lifecycle	TC-SPR-01; start and complete transitions in `sprintLifecycle.test.js`	Pass
PMS-F-011	Burndown chart	TC-BURN-01	Pass
PMS-F-012	KPI dashboard	TC-DASH-01	Pass
PMS-F-013	Team workload	TC-WORK-01	Pass
PMS-F-014	Audit trail	TC-LOG-01	Pass
PMS-F-015	Supabase synchronisation	TC-SUPA-01, TC-SET-01	**Blocked** in the testing phase (Section 6); **Pass** on 11-11-2026 on a disposable Supabase database (Section 6.1)
PMS-F-016	Offline mode	TC-OFF-01, TC-AVAIL-01	Pass
PMS-F-017	Filtering	TC-FILT-01	**Fail** in the main run (DEF-04, low); **Pass** on retest of the fix, PR #98
PMS-F-018	CI/CD pipeline	GitHub Actions history (Unit Test Report, Section 9)	Pass
PMS-NF-001	Performance, under 50 ms	TC-PERF-01 (p95 6.6 ms), TC-LOAD-01 (p95 8.9 ms with 106 tasks)	Pass
PMS-NF-002	Availability through offline fallback	TC-AVAIL-01	Pass
PMS-NF-003	Usability: contrast, Lighthouse score above 90	No case in the specification; keyboard and screen-reader behaviour covered by `accessibility.test.jsx`; Lighthouse accessibility audit of the sign-in page and the board (Section 12.3)	**Not measured** in the testing phase; **Pass** on 11-11-2026: Lighthouse accessibility score 100 on both screens (Section 12.3)
PMS-NF-004	Reliability: zero data loss across reloads and tab closures	TC-OFF-01, TC-RECOV-01, refresh steps in TC-KAN-02 and TC-TASK-01; TC-SESS-01 (supplementary, simulated Supabase, Section 12.2)	Pass
PMS-NF-005	Maintainability: zero ESLint warnings, build always passes	`eslint --max-warnings=0`: 0 errors, 0 warnings; `npm run build`: succeeded (143 modules) on the same commit	Pass
PMS-SR-001	Row Level Security	SEC-RLS	**Blocked** in the testing phase (Section 6); **Pass** on 11-11-2026 on a disposable Supabase database (Section 6.1)
PMS-SR-002	No hardcoded credentials	SEC-KEYS	Pass
PMS-SR-003	HTTPS / TLS 1.3	TC-SEC-01	**Blocked** in the testing phase (Section 6); **Pass** on 11-11-2026 on a disposable Supabase database (Section 6.1)
PMS-SR-004	XSS input sanitisation	SEC-XSS, SEC-FUZZ	Pass
PMS-SR-005	Dependency audit in CI	Dependency Security Audit workflow: 10 of 10 runs passed; `npm audit --audit-level=high`: no high or critical vulnerabilities	Pass
(Test Plan only)	In-app documentation viewer	TC-DOCS-01	Pass
(Test Plan only)	Cross-browser support	TC-COMPAT-01	Pass
(Test Plan only)	100+ task records	TC-LOAD-01	Pass

SRS ID	Requirement	Verified by	Result
(Test Plan only)	Session token handling	SEC-TOKEN	Pass

Summary against the SRS:

- Functional (PMS-F-001 to 018): in the main run 16 pass, 1 failed with a low-severity defect (now fixed and retested) and 1 was blocked. The blocked one, PMS-F-015, passed on 11-11-2026, so all 18 now pass.
- Non-functional (PMS-NF-001 to 005): 4 pass and 1 was not measured in the testing phase. PMS-NF-003 was measured on 11-11-2026 and passed, so all 5 now pass.
- Security (PMS-SR-001 to 005): 3 pass and 2 were blocked in the testing phase. Both passed on 11-11-2026, so all 5 now pass.

Every SRS requirement has at least one test case or piece of objective evidence.

9. Cross-Browser Compatibility (TC-COMPAT-01)

TC-KAN-01, TC-KAN-02 and TC-TASK-01 were run in each browser in a fresh context, and the position and width of every column were compared with Chrome.

Browser	Version	TC-KAN-01	TC-KAN-02	TC-TASK-01	Layout difference from Chrome
Google Chrome	157.0.8103.38	Pass	Pass	Pass	–
Microsoft Edge	156.0.4314.58	Pass	Pass	Pass	0 px
Firefox (Playwright build)	142.0.1	Pass	Pass	Pass	0 px

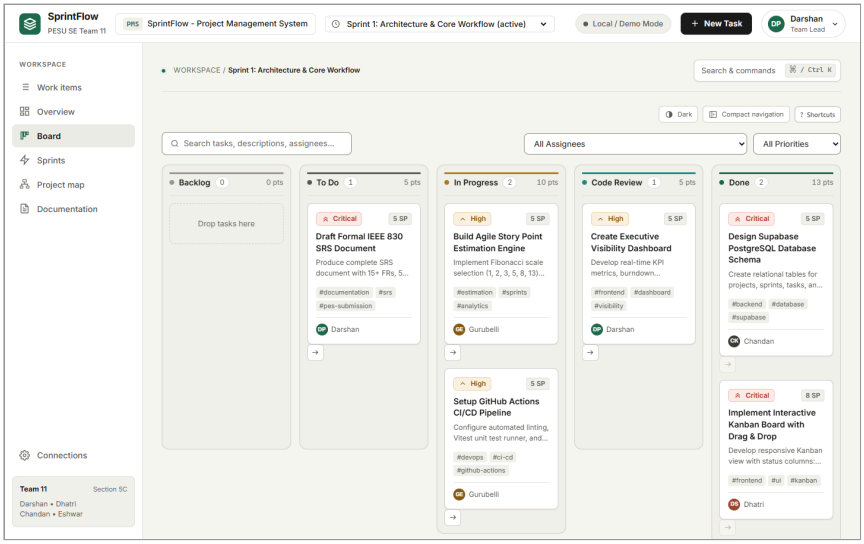


Figure 3: Chrome 157

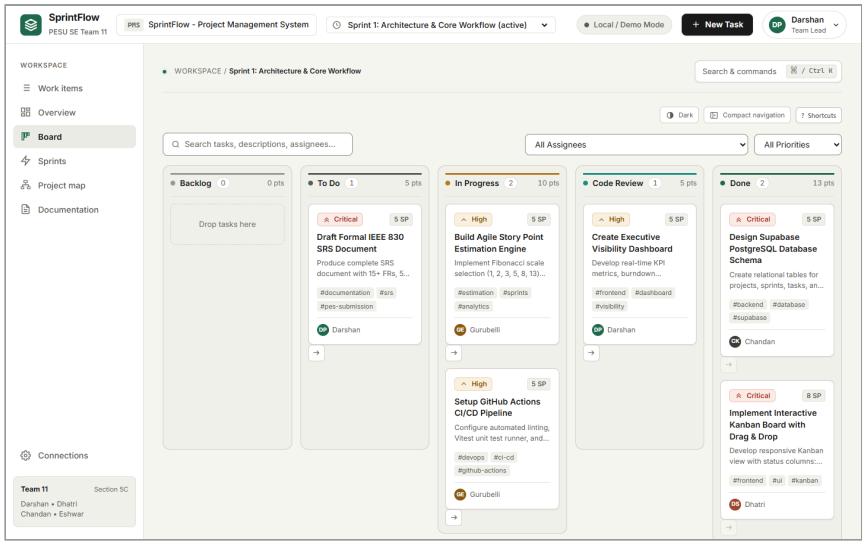


Figure 4: Edge 156

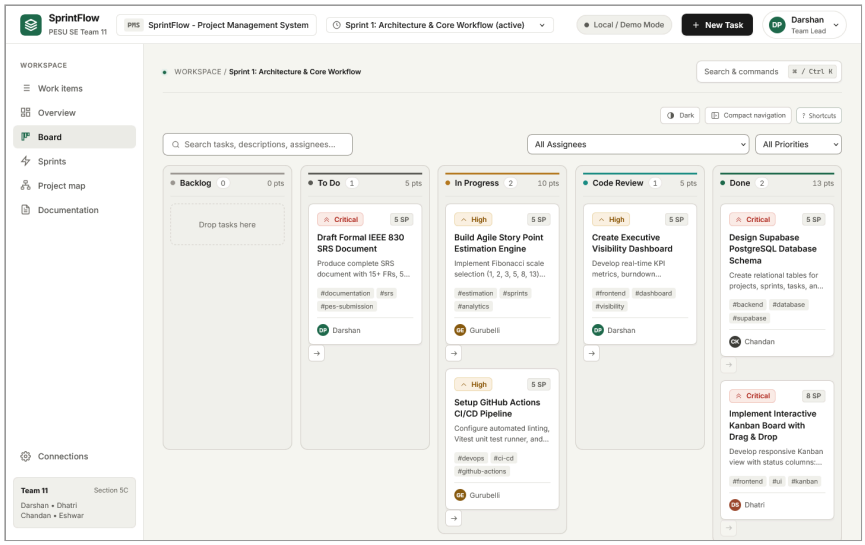


Figure 5: Firefox 142

The board after sign-in in each browser, same build and seed data: five columns and six cards in the same positions. Captured again on 06-11-2026 after the run, because the first Chrome capture was taken while the board was still fading in. Full-size images are in `docs/test-reports/evidence/`.

10. Security Validation (STP Section 5.4)

STP check	Case	Method	Result
XSS input sanitisation	SEC-XSS	`<img onerror>`, `<script>` and `<svg onload>` payloads in a task title, task description and sprint goal; a probe variable shows whether any handler ran	Pass: nothing ran; no injected element in the page
Row Level Security	SEC-RLS	Cross-project REST call from a second account	Blocked in the testing phase (Section 6), policies verified statically by `databaseSecurity.test.js`; passed on 11-11-2026 on a disposable Supabase database (Section 6.1)
TLS 1.3 enforcement	TC-SEC-01	Inspect live traffic	Blocked in the testing phase (Section 6), and the client refuses `http://` project URLs; passed on 11-11-2026 (Section 6.1)

STP check	Case	Method	Result
Session token handling	SEC-TOKEN	Scan localStorage and sessionStorage after sign-in	Pass: neither the password nor any JWT is stored in zero-configuration mode. Rejection of expired tokens was not tested in the testing phase; it passed on 11-11-2026 as SEC-EXPIRED (Section 12.3)
API key exposure	SEC-KEYS	Scan the production bundle for complete JWTs and Supabase secret keys	Pass: none in `dist/assets`
Input validation fuzzing	SEC-FUZZ	5,000-character task title; Unicode, quotes, `&`, SQL-like text and emoji (the title was the only field fuzzed)	Pass: title bounded to 120 characters with no crash; special characters stored and shown as typed

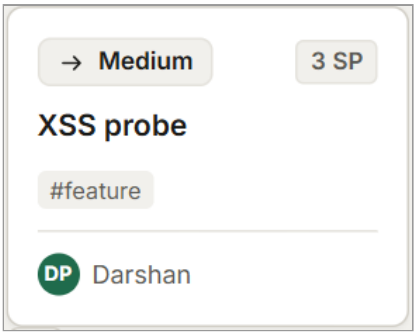


Figure 6: SEC-XSS: the tags in the title were removed, only the text "XSS probe" is shown, and the image handler never ran.

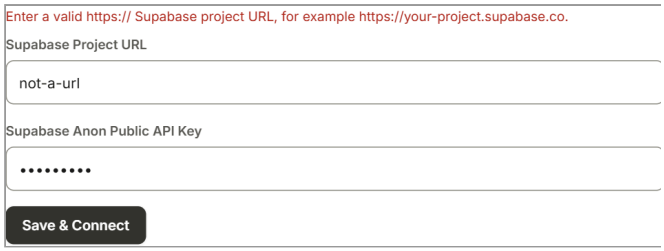


Figure 7: TC-SET-01 step 3: a malformed project URL is refused with a clear message.

11. Exit Criteria Assessment (STP Section 5.3)

Exit criterion	Status	Detail
100% of planned test cases executed	Met on 11-11-2026	23 of the 26 specified cases ran in full during the testing phase. The other three, TC-SUPA-01, TC-SET-01 (steps 1–2) and TC-SEC-01, needed a Supabase project (#96). They were run on 11-11-2026 on a disposable Supabase database and passed (Section 6.1), so all 26 have now run in full.
0 critical or high-severity defects open	Met	Two defects were found, neither critical nor high, and both are fixed and retested: DEF-04 (PR #98) and DEF-05 (PR #101).
All functional requirements verified against acceptance criteria	Met on 11-11-2026	17 of 18 SRS functional requirements passed in the testing phase once the DEF-04 fix is counted. PMS-F-015 (Supabase sync) was blocked, and passed TC-SUPA-01 and TC-SET-01 on 11-11-2026 (Section 6.1).
All non-functional requirements measured and within threshold	Met on 11-11-2026	4 of 5 SRS non-functional requirements were measured in the testing phase, all within threshold. PMS-NF-003 had no test case; on 11-11-2026 Lighthouse gave an accessibility score of 100 on the sign-in page and on the board, against a target above 90 (Section 12.3). The STP lists six non-functional requirements because it also counts network recovery (PMS-NF-006 in its numbering), tested by TC-RECOV-01, which passed.
RTM shows complete coverage	Met	Every SRS requirement maps to a test case or objective evidence (Section 8).
Test Summary Report reviewed and approved	Met on 11-11-2026	There was no separate Test Summary Report at the end of the testing phase. One was added in PR #107 (`docs/test-reports/TEST_SUMMARY_REPORT.md`), which the Test Architect, Darshan P Pawar, approved and merged on 11-11-2026.

The two criteria that were open at the end of the testing phase were closed on 11-11-2026, before submission. The Lighthouse measurement is not yet a case in the Test Case Specification; we recommend adding it when the specification is updated (Part 3, item 1).

12. Regression on the Final Build

After the main run, five pull requests were merged into `main` on 06-11-2026: #97 (the first version of these reports and the system test script), #98 (the DEF-04 fix), #99 (Dhatri's second-machine pre-check), #101 (the DEF-05 fix, a tooltip warning and the CI Node.js version) and #85 (verified cloud sessions, merged after review fixes). Because #85 changes how sign-in works, we ran the whole suite again on the final `main`.

Run	Build	Browsers	Pass	Fail	Blocked
Main run (Sections 3 to 11)	`f9e0727`	Chrome; Edge and Firefox for TC-COMPAT-01	28	1	4
DEF-04 retest (Section 5.2)	`e8f5905` (PR #98)	Chrome	29	0	4
Final regression, 06-11-2026, 23:05 IST	`2b8fdbc` (final `main`)	Chrome; Edge and Firefox for TC-COMPAT-01	29	0	4
Pre-submission regression, 11-11-2026, 12:08 IST (Section 12.3)	`970c656` (final `main`, GitHub Actions)	Chrome; Edge and Firefox for TC-COMPAT-01	29	0	4
Cloud checks in the same run, on a disposable Supabase database (Sections 6.1 and 12.3)	`970c656`	Chrome and direct API calls	7	0	0

In the final regression of 06-11-2026 the results matched the main run case for case, except that TC-FILT-01 now passed. TC-COMPAT-01 again found a 0 px layout difference between the three browsers, and the same four cloud cases stayed blocked until 11-11-2026 (Section 6.1). On the same commit all 259 unit tests passed (Unit Test Report, Section 10).

12.1 DEF-05 retest

PR #101 makes the panel say "task" for a count of one and adds a unit test for it. On the final build Dhatri and Chandan now read "1 task • 8 story points assigned" and "1 task • 5 story points assigned", while members with two tasks still read "2 tasks". TC-WORK-01 passes, and issue #100 was closed by the merge.

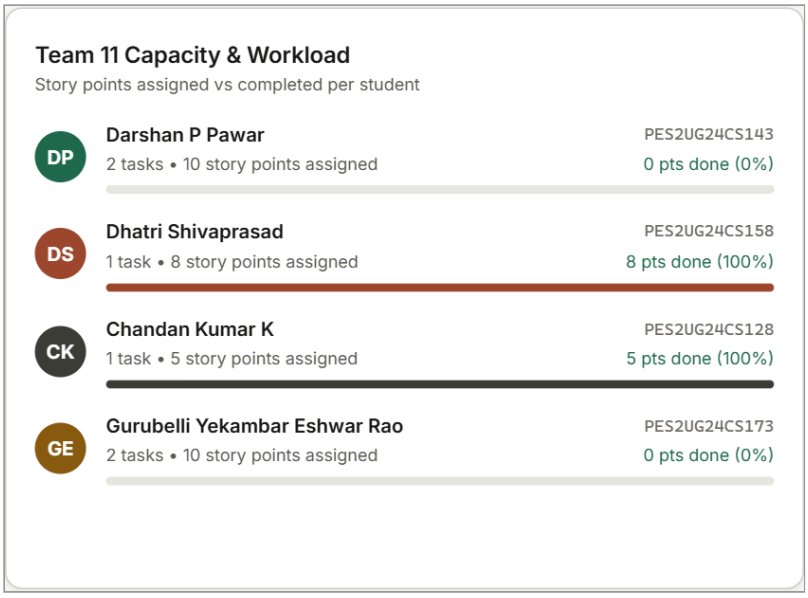


Figure 8: DEF-05 retest on the final build: "1 task" for a member with one task.

12.2 Supplementary case TC-SESS-01: a network failure must not sign out a member

PR #85 re-checks the cloud session whenever Supabase reports an auth event, for example when the browser tab regains focus. While reviewing it on 03-11-2026 (review comment on PR #85) we found that a failed check signed the member out even when only the network had failed, which threw away any task being edited. That is a data-loss risk under PMS-NF-004. The fix keeps the member on network errors and signs out only when Supabase rejects the

session. No Supabase test project was available at the time (Section 6), so we ran this case in Chrome against a simulated Supabase server: Playwright answered the app's Supabase requests, and no real database was used.

Step	Expected result	Before the fix	Final build (`2b8fdbc`)
1. Sign in to the cloud project	The workspace opens	Pass	Pass
2. Type a task title; the session check then fails because the network is down	Still signed in; the dialog and the typed title remain	**Fail**: sent to the sign-in page, draft lost	Pass
3. The network comes back	The session is checked again; still signed in	Not reached	Pass
4. Supabase rejects the session (401)	Signed out with "Your cloud session has ended"	Not reached	Pass

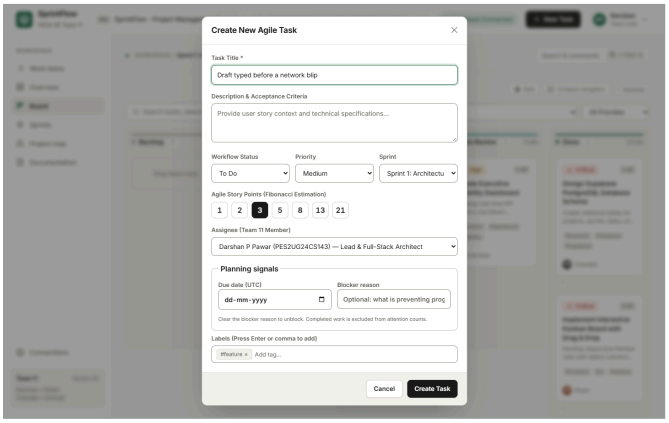


Figure 9: Step 2 on the final build: still signed in, the draft is kept.

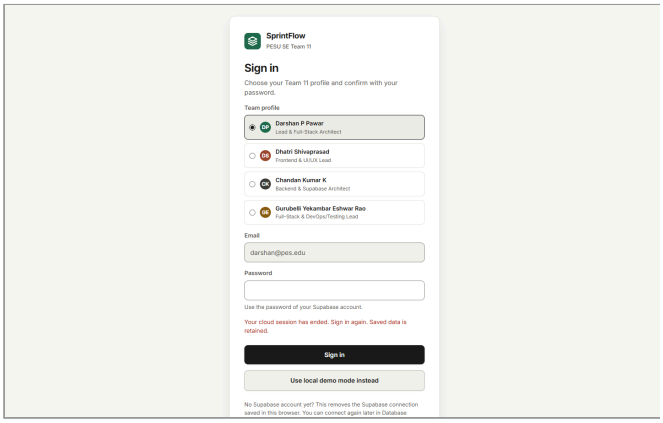


Figure 10: Step 4: a rejected session signs the member out with a clear message.

The same logic is covered by 15 unit tests in `src/tests/sessionVerification.test.js`, which fail on the code before the fix and pass after it. This case does not replace TC-SUPA-01 on a real Supabase database, which passed on 11-11-2026 (Section 6.1).

12.3 Pre-submission regression on 11-11-2026

Four more pull requests were merged before submission: #103, #104, #105 and #107 (Unit Test Report, Section 10). #107 also automated the cloud cases and an accessibility audit. On 11-11-2026 the "Sprint 1-2 Testing Evidence" workflow ran everything on the final `main` (commit `970c656`) in GitHub Actions run 42973669659, on Ubuntu 24.04 with Chrome 157.0.8103.50, Edge 157.0.4341.46 and Firefox 142.0.1.

- **Browser suite:** 29 of 33 cases passed and none failed, the same result as the final regression of 06-11-2026. The 4 blocked cases are the ones that need a database. TC-COMPAT-01 passed in all three browsers (9 of 9 runs).
- **Cloud cases:** all four cases blocked in the browser suite passed on the disposable Supabase database (Section 6.1), so all 33 specified cases now pass. Three further checks ran on the same database and passed (table below).
- **Accessibility (PMS-NF-003):** Lighthouse 13.6.0 in Chrome, desktop layout at 1440 x 900, gave an accessibility score of 100 on the sign-in page and 100 on the board, with no failed audits. The SRS target is a score above 90. Only these two screens were audited.
- **TC-SESS-01:** the four steps of Section 12.2 were repeated against the simulated Supabase server, and all 4 passed.

Check	What it shows	Result
SEC-EXPIRED	A correctly signed but expired session token is refused by Supabase Auth and by the REST API, and a malformed token is refused too.	**Pass**
SEC-AUDIT	Writing an audit record in another member's name, or changing or deleting audit history, is refused.	**Pass**
CLOUD-RECOVERY	A task created while the browser is offline stays visible, reaches PostgreSQL once the connection returns, and is stored exactly once.	**Pass**

13. Conclusion

In zero-configuration mode SprintFlow behaves as specified in Chrome, and the core board and task workflows behave the same in Edge and Firefox (TC-COMPAT-01). Every specified workflow passed: creating, editing and deleting tasks, drag-and-drop, one-click advance, estimation, sprint planning, the burndown chart, the dashboard, the audit trail, offline use and recovery. Drag-and-drop rendered within 9 ms at the 95th percentile even with 106 tasks on the board, against a 50 ms budget. No XSS payload ran and no credential was exposed in the bundle. Testing found two defects, a low-severity usability defect (DEF-04) and a cosmetic wording defect (DEF-05). Both were fixed, retested and merged on 06-11-2026, and a full regression run on the final build in all three browsers passed with no failures. The four cloud-database cases that were blocked in the testing phase passed on 11-11-2026 on a disposable Supabase database, and Lighthouse rated the accessibility of both audited screens 100, so all 33 cases and every SRS requirement in Section 8 now pass. A check of the team's hosted Supabase project itself was outside this test run.

Appendix A: Detailed Results

Each case from the main run with every expected result that was checked and the actual result the browser showed.

TC-AUTH-01: Authenticate and establish session (Pass)

SRS: PMS-F-001 · Test Plan: PMS-F-001 · Browser: Chrome

Expected result checked	Actual result	Verdict
Navigation bar shows the profile name	Darshan	Pass
Navigation bar shows the role	Team Lead	Pass
Session persists after refresh (no sign-in view)	signed in after reload	Pass

TC-PROJ-01: Show project metadata (supplementary) (Pass)

SRS: PMS-F-002 · Test Plan: – · Browser: Chrome

Expected result checked	Actual result	Verdict
Top bar shows the project key and name	SprintFlow PESU SE Team 11 PMS SprintFlow - Project Management System Sprint 1: Architecture & Core Workflow (active) Sp…	Pass
Top bar shows the active sprint	Sprint 1: Architecture & Core Workflow (active)	Pass
Dashboard shows the project key, name and date range	2026-09-01 → 2026-10-30	Pass

TC-TASK-01: Create a task with complete metadata (Pass)

SRS: PMS-F-006 · Test Plan: PMS-F-002 · Browser: Chrome

Expected result checked	Actual result	Verdict
Card shows the title	High 5 SP Implement burndown chart #feature DS Dhatri	Pass
Card shows a High priority badge	High 5 SP Implement burndown chart #feature DS Dhatri	Pass
Card shows a 5-point estimate	High 5 SP Implement burndown chart #feature DS Dhatri	Pass
Card shows the assignee	High 5 SP Implement burndown chart #feature DS Dhatri	Pass
With Backlog chosen, the card appears at the top of the Backlog column	Implement burndown chart (Backlog)	Pass
Cards survive a page refresh	both present after reload	Pass

Notes: With the default Workflow Status the card lands in "To Do". SPRINT1_DEVIATIONS.md records that a new task starts in To Do unless another status is chosen; the specification text still says Backlog.

TC-TASK-02: Assign Critical priority (Pass)

SRS: PMS-F-006 · Test Plan: PMS-F-016 · Browser: Chrome

Expected result checked	Actual result	Verdict
Card shows a Critical badge	Critical	Pass
Badge carries a text label, not colour alone	Critical	Pass

TC-TASK-03: Reject a task with an empty title (Pass)

SRS: PMS-F-006 · Test Plan: PMS-F-002 · Browser: Chrome

Expected result checked	Actual result	Verdict
Empty title is rejected with a validation message	browser validation: "Please fill out this field."	Pass
Single-space title is rejected with a validation message	inline message: "Enter a task title and choose a valid priority."	Pass
No task is created in either case	6 cards before and after	Pass

TC-LABEL-01: Add labels to a task (supplementary) (Pass)

SRS: PMS-F-007 · Test Plan: – · Browser: Chrome

Expected result checked	Actual result	Verdict
Labels show as separate pills, lower-cased and de-duplicated	#feature #ui #api	Pass
Labels survive a page refresh	#feature #ui #api	Pass

TC-KAN-01: Render the five-column board (Pass)

SRS: PMS-F-003 · Test Plan: PMS-F-003 · Browser: Chrome

Expected result checked	Actual result	Verdict
Exactly five columns in lifecycle order	Backlog, To Do, In Progress, Code Review, Done	Pass
Backlog shows its task count	count badge 0, cards 0	Pass
To Do shows its task count	count badge 1, cards 1	Pass
In Progress shows its task count	count badge 2, cards 2	Pass
Code Review shows its task count	count badge 1, cards 1	Pass
Done shows its task count	count badge 2, cards 2	Pass

TC-KAN-02: Transition a task by drag-and-drop (Pass)

SRS: PMS-F-004 · Test Plan: PMS-F-004 · Browser: Chrome

Expected result checked	Actual result	Verdict
Card renders in In Progress on release	In Progress	Pass
Source column count decreases by one	1 → 0	Pass
Target column count increases by one	2 → 3	Pass
New state survives a refresh	In Progress	Pass

TC-KAN-03: Advance a task by single action (Pass)

SRS: PMS-F-005 · Test Plan: PMS-F-005 · Browser: Chrome

Expected result checked	Actual result	Verdict
Precondition: task is in To Do	To Do	Pass
Task moves to In Progress	In Progress	Pass
A task in Done cannot advance further	0 enabled advance controls in Done	Pass

TC-KAN-04: Cancel a drag outside any column (Pass)

SRS: PMS-F-004 · Test Plan: PMS-F-004 · Browser: Chrome

Expected result checked	Actual result	Verdict
A drag was actually started	1 dragstart, 0 drop events	Pass
Card returns to its original column	In Progress → In Progress	Pass
Status is unchanged after refresh	In Progress	Pass

TC-FILT-01: Filter by priority (Fail)

SRS: PMS-F-017 · Test Plan: PMS-F-011 · Browser: Chrome

Expected result checked	Actual result	Verdict
Only High priority tasks are visible	3 visible: high, high, high	Pass
No task matches the combined filters	0 cards visible	Pass
An explicit empty state names the active filters	five columns each show "Drop tasks here"; no message names the active filters	Fail
The empty state offers to clear the filters	"Reset Filters" button shown	Pass

TC-EST-01: Assign Fibonacci story points (Pass)

SRS: PMS-F-008 · Test Plan: PMS-F-006 · Browser: Chrome

Expected result checked	Actual result	Verdict
Only the seven Fibonacci values are offered	1, 2, 3, 5, 8, 13, 21	Pass
No free numeric estimate field exists	0 numeric inputs	Pass
Estimate 1 persists on the card	1 SP	Pass
Estimate 2 persists on the card	2 SP	Pass
Estimate 3 persists on the card	3 SP	Pass
Estimate 5 persists on the card	5 SP	Pass
Estimate 8 persists on the card	8 SP	Pass
Estimate 13 persists on the card	13 SP	Pass
Estimate 21 persists on the card	21 SP	Pass
The final estimate is reflected in sprint capacity	33 → 49 (task 5 → 21)	Pass

TC-EST-02: Calculate sprint capacity (Pass)

SRS: PMS-F-009 · Test Plan: PMS-F-008 · Browser: Chrome

Expected result checked	Actual result	Verdict
Precondition: the new sprint has no tasks	0	Pass
Capacity reports 16 points for tasks of 3, 5 and 8	16	Pass
Capacity is recomputed to 11 after removing the 5-point task	11	Pass

TC-SPR-01: Create a sprint (Pass)

SRS: PMS-F-010 · Test Plan: PMS-F-007 · Browser: Chrome

Expected result checked	Actual result	Verdict
Sprint appears in the sprint list	1	Pass
Sprint shows its goal	PLANNED 2026-11-06 → 2026-11-19 Sprint 3 View sprint Start Sprint Delete Sprint Sprint Goal: Deliver estimation engine 0 Planned Points 0 Completed 0 Remaining …	Pass
Sprint shows its two-week date range	2026-11-06 → 2026-11-19	Pass
Sprint is available as a task assignment target	Backlog (No Sprint) | Sprint 1: Architecture & Core Workflow | Sprint 2: Analytics & Release Candidate | Sprint 3	Pass

TC-BURN-01: Update the burndown chart (Pass)

SRS: PMS-F-011 · Test Plan: PMS-F-009 · Browser: Chrome

Expected result checked	Actual result	Verdict
Precondition: today has a recorded actual value	20	Pass
Task moved to Done	Done	Pass
Actual line falls by exactly the completed points	20 → 15 (task 5 SP)	Pass
Ideal line is unchanged	19 ideal values compared	Pass

Notes: Read from the chart's "Daily burndown values" table, which lists the same series the SVG draws.

TC-DASH-01: Display KPI metrics (Pass)

SRS: PMS-F-012 · Test Plan: PMS-F-010 · Browser: Chrome

Expected result checked	Actual result	Verdict
Progress, velocity and workload are shown	total 33, delivered 13, completion 39%, remaining 20	Pass
Total matches the board	board 33 pts, dashboard 33	Pass
Delivered velocity matches the Done column	Done 13 pts, dashboard 13	Pass
Progress equals completed points / total points	13/33 = 39.4%, shown 39%	Pass

TC-WORK-01: Show workload distribution (Pass)

SRS: PMS-F-013 · Test Plan: PMS-F-017 · Browser: Chrome

Expected result checked	Actual result	Verdict
Tasks are assigned across at least three members	10, 8, 5, 10	Pass
Member totals sum to the sprint total	10 + 8 + 5 + 10 = 33; sprint total 33	Pass

TC-DOCS-01: Render in-app documentation (Pass)

SRS: not in SRS · Test Plan: PMS-F-018 · Browser: Chrome

Expected result checked	Actual result	Verdict
"Functional Requirements" tab renders readable content	3172 characters	Pass
"Non-Functional" tab renders readable content	1093 characters	Pass
"Traceability Matrix" tab renders readable content	947 characters	Pass
Documentation has a readable heading	Software Requirements Specification (SRS) & RTM	Pass

TC-SUPA-01: Persist to Supabase PostgreSQL (Blocked)

SRS: PMS-F-015 · Test Plan: PMS-F-013 · Browser: Chrome

Blocked: Precondition not met: needs a Supabase test project with database/schema.sql (and the 20261103 migration) applied.

TC-OFF-01: Persist offline without configuration (Pass)

SRS: PMS-F-016 · Test Plan: PMS-F-014 · Browser: Chrome

Expected result checked	Actual result	Verdict
Both tasks are restored after reopening the tab	2 of 2 present	Pass
Interface indicates offline (localStorage) persistence	Local / Demo Mode	Pass

TC-SET-01: Switch data mode from settings (Blocked)

SRS: PMS-F-015 · Test Plan: PMS-F-015 · Browser: Chrome

Blocked: Steps 1-2 need valid credentials for a provisioned Supabase test project; only step 3 (negative case) was executed.

Expected result checked	Actual result	Verdict
Step 3: a malformed URL is rejected with a clear message	Enter a valid https:// Supabase project URL, for example https://your-project.supabase.co.	Pass
Step 3: the previous (offline) configuration is retained	Local / Demo Mode	Pass

TC-LOG-01: Record an audit trail for mutations (Pass)

SRS: PMS-F-014 · Test Plan: PMS-F-012 · Browser: Chrome

Expected result checked	Actual result	Verdict
Exactly three new audit entries	4 → 7	Pass
Entries are CREATE, UPDATE, DELETE in chronological order	CREATE, UPDATE, DELETE	Pass
Each entry carries a timestamp	2026-11-06T12:08:22.068Z, 2026-11-06T12:08:22.563Z, 2026-11-06T12:08:23.106Z	Pass
Timestamps are in chronological order	non-decreasing	Pass
Each entry names the acting user and the task	CREATE Darshan P Pawar created task TC-LOG-01 audit probe → To Do || UPDATE Darshan P Pawar updated task TC-LOG-01 audit probe (renamed) || DELETE Darshan P Pawar deleted task TC-LOG-01 audit probe (renamed)…	Pass

TC-PERF-01: Drag-and-drop transition under 50 ms (Pass)

SRS: PMS-NF-001 · Test Plan: PMS-NF-001 · Browser: Chrome

Expected result checked	Actual result	Verdict
Drop-to-render p95 under 50 ms	6.6 ms	Pass

Notes: 10 measured drops after 2 warm-ups: p50 2.3 ms, p95 6.6 ms, max 6.6 ms.

TC-AVAIL-01: Stay usable with the network disabled (Pass)

SRS: PMS-NF-002 · Test Plan: PMS-NF-002 · Browser: Chrome

Expected result checked	Actual result	Verdict
Application stays usable with the network disabled	task created; Overview, Sprints, Work items and Board opened	Pass
Application reports offline (local) mode	Local / Demo Mode	Pass

TC-COMPAT-01: Core workflows on Chrome, Edge and Firefox (Pass)

SRS: not in SRS · Test Plan: PMS-NF-003 · Browser: Chrome, Edge and Firefox

Expected result checked	Actual result	Verdict
chrome: TC-KAN-01, TC-KAN-02 and TC-TASK-01 pass	TC-KAN-01 pass, TC-KAN-02 pass, TC-TASK-01 pass	Pass
edge: TC-KAN-01, TC-KAN-02 and TC-TASK-01 pass	TC-KAN-01 pass, TC-KAN-02 pass, TC-TASK-01 pass	Pass
firefox: TC-KAN-01, TC-KAN-02 and TC-TASK-01 pass	TC-KAN-01 pass, TC-KAN-02 pass, TC-TASK-01 pass	Pass
edge: column layout matches chrome within 4 px	max drift 0 px	Pass
firefox: column layout matches chrome within 4 px	max drift 0 px	Pass

Notes: Browser versions: chrome 157.0.8103.38, edge 156.0.4314.58, firefox 142.0.1.

TC-SEC-01: Encrypted transport to Supabase (Blocked)

SRS: PMS-SR-003 · Test Plan: PMS-NF-004 · Browser: Chrome

Blocked: Inspecting live HTTPS traffic needs a connected Supabase project. Client-side enforcement was verified instead.

Expected result checked	Actual result	Verdict
A plain http:// project URL is refused	Enter a valid https:// Supabase project URL, for example https://your-project.supabase.co.	Pass

TC-LOAD-01: Board with 100+ tasks (Pass)

SRS: not in SRS · Test Plan: PMS-NF-005 · Browser: Chrome

Expected result checked	Actual result	Verdict
All seeded tasks render on the board	106 cards	Pass
Every seeded task is accessible as a control	100 task controls	Pass
Drag-and-drop stays responsive (p95 under 50 ms with 106 tasks)	8.9 ms	Pass

Notes: Board ready 162 ms after navigation with 106 tasks; drop-to-render p95 8.9 ms over 5 drops.

TC-RECOV-01: No data loss across disconnect and reconnect (Pass)

SRS: PMS-NF-004 · Test Plan: PMS-NF-006 · Browser: Chrome

Expected result checked	Actual result	Verdict
No task is lost across disconnect and reconnect	2 of 2 present	Pass

Notes: Zero-config mode. The cloud retry queue across a reconnect is covered by unit tests (persistenceRecovery).

SEC-XSS: XSS input sanitisation (Pass)

SRS: PMS-SR-004 · Test Plan: STP 5.4 · Browser: Chrome

Expected result checked	Actual result	Verdict
No injected script or handler executes	window.__sfXss never set	Pass
Input is not rendered as markup (tags removed)	0 injected elements in cards	Pass

SEC-RLS: Row Level Security on direct API calls (Blocked)

SRS: PMS-SR-001 · Test Plan: STP 5.4 · Browser: Chrome

Blocked: Needs a live Supabase project to send cross-project API calls. The policies in database/schema.sql are checked statically by src/tests/databaseSecurity.test.js.

SEC-TOKEN: Session token handling (Pass)

SRS: not in SRS · Test Plan: STP 5.4 · Browser: Chrome

Expected result checked	Actual result	Verdict
The password is not kept in browser storage	not found	Pass
No access token is kept in zero-config mode	no JWT in storage	Pass

Notes: Cloud JWT expiry and rejection need a live Supabase project (see TC-SEC-01).

SEC-KEYS: API key exposure in the client bundle (Pass)

SRS: PMS-SR-002 · Test Plan: STP 5.4 · Browser: none (static scan of the build)

Expected result checked	Actual result	Verdict
No complete JWT (anon or service_role) is embedded in the bundle	0 found in 2 files	Pass
No Supabase secret key is embedded in the bundle	0 found	Pass

Notes: Static scan of the production bundle in dist/assets.

SEC-FUZZ: Input validation fuzzing (Pass)

SRS: PMS-SR-004 · Test Plan: STP 5.4 · Browser: Chrome

Expected result checked	Actual result	Verdict
An over-long title is bounded, not crashing the app	120 characters stored	Pass
Special characters are stored and shown as typed	Ω ✓ "quotes" & ampersand; DROP TABLE tasks; -- 🚀	Pass

Appendix B: Screenshot Evidence

These screenshots were captured on 06-11-2026 after the main run by repeating the main steps of each case by hand in a fresh Chrome session on the same build (`f9e0727`).

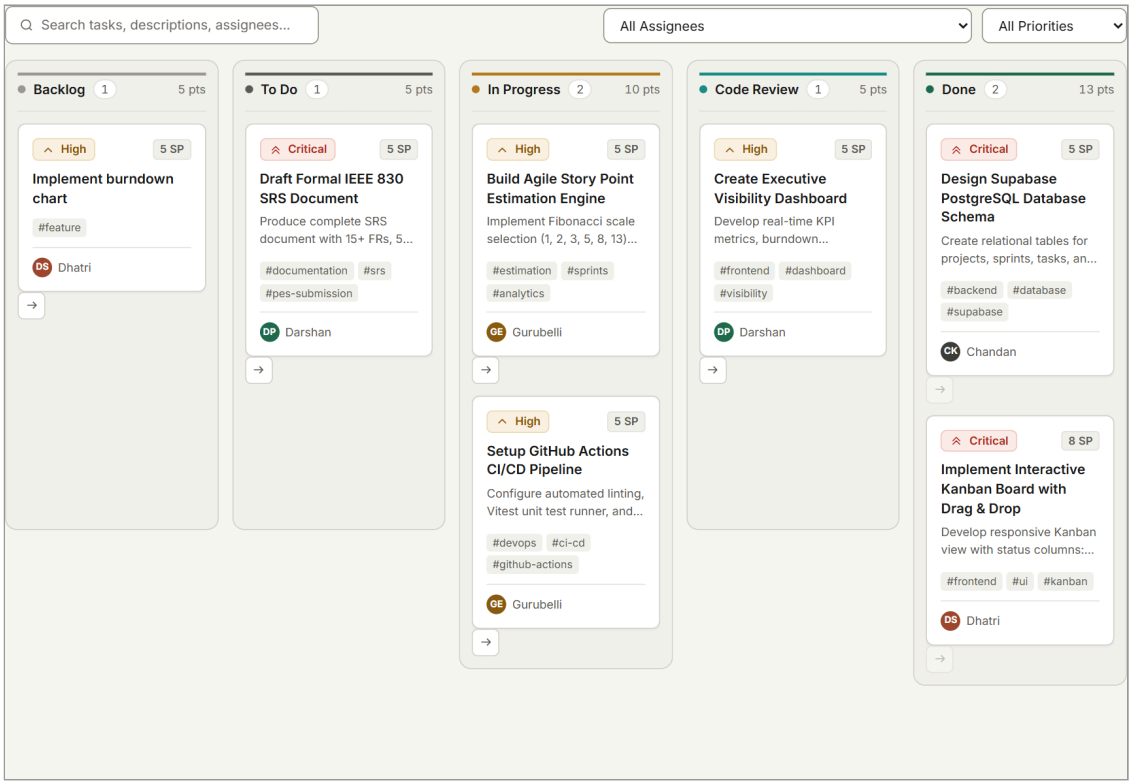


Figure 11: TC-TASK-01, repeated by hand: "Implement burndown chart" (High, 5 SP, Dhatri), created with Backlog chosen as its status, at the top of Backlog after a page refresh.

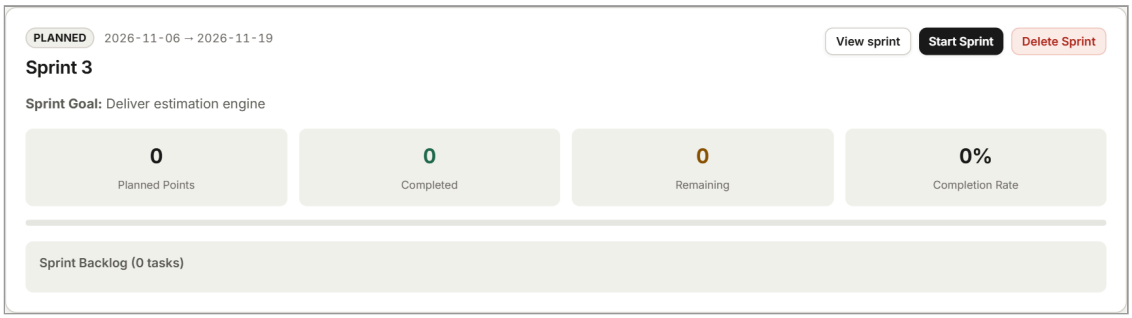


Figure 12: TC-SPR-01: Sprint 3 with its goal and two-week date range, in the Planned state.

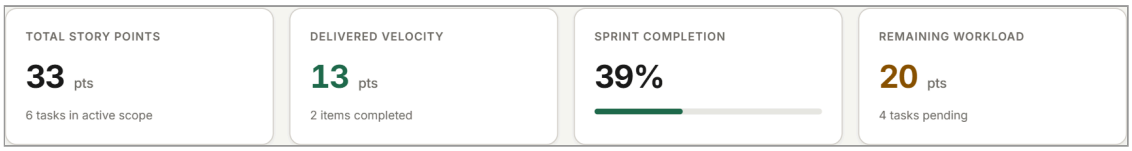


Figure 13: TC-DASH-01: KPI cards (total 33 points, 13 delivered, 39% complete, 20 remaining).

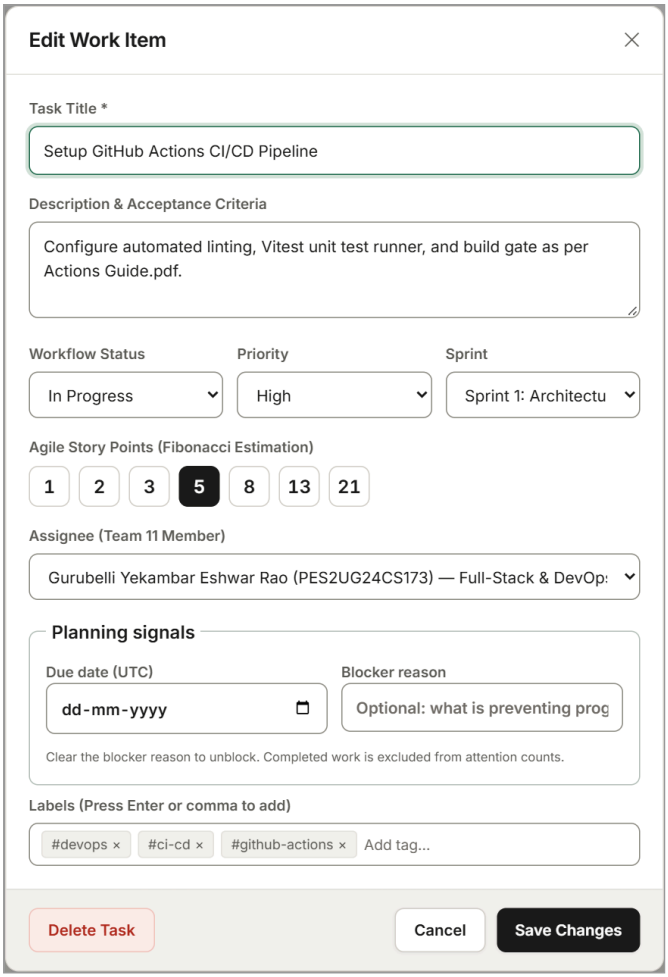


Figure 14: TC-EST-01: the task dialog offers only 1, 2, 3, 5, 8, 13 and 21.

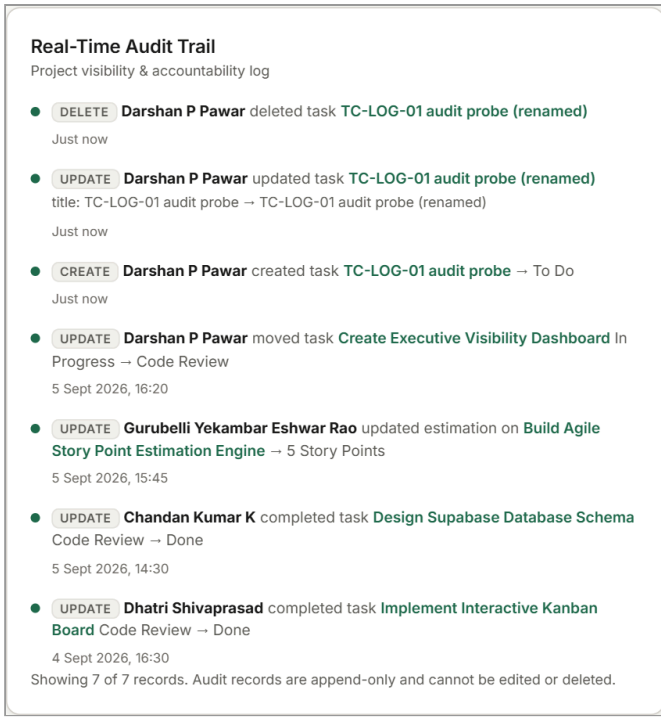


Figure 15: TC-LOG-01: CREATE, UPDATE and DELETE records with the user, task and time.

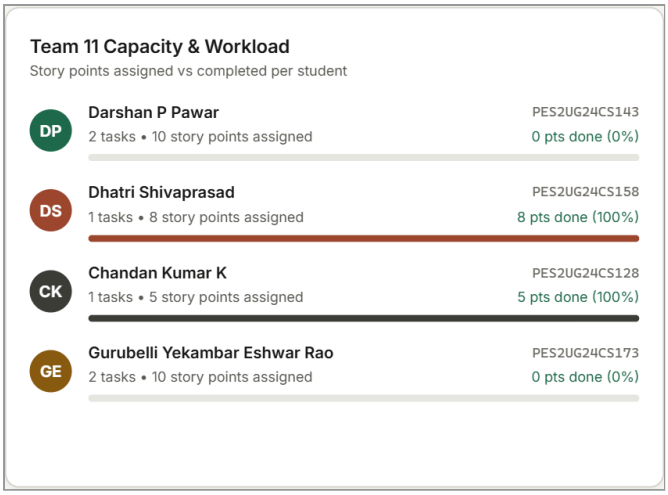
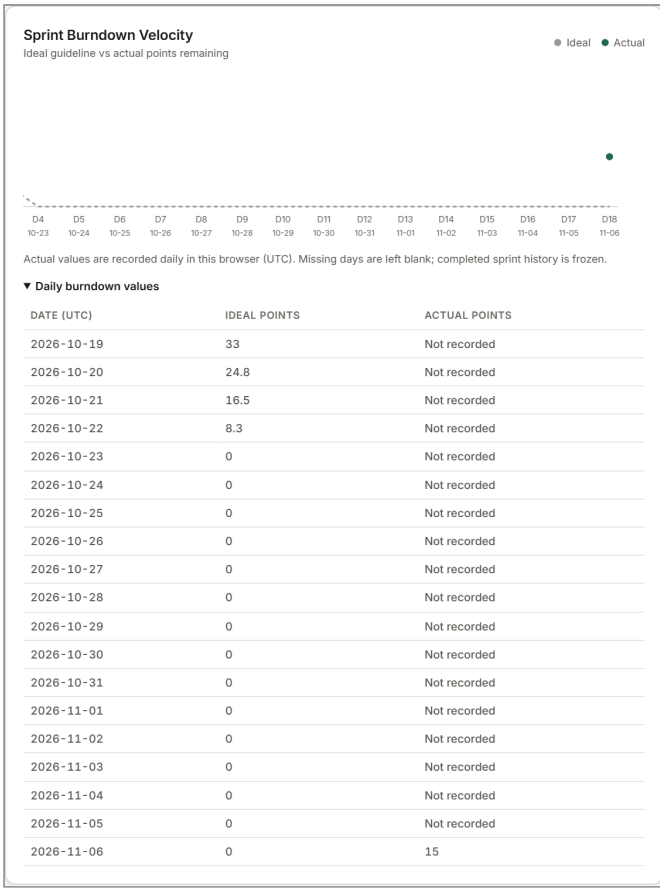


Figure 16: TC-WORK-01: points per member add up to the sprint total (note "1 tasks", DEF-05).



DATE (UTC)	IDEAL POINTS	ACTUAL POINTS
2026-10-19	33	Not recorded
2026-10-20	24.8	Not recorded
2026-10-21	16.5	Not recorded
2026-10-22	8.3	Not recorded
2026-10-23	0	Not recorded
2026-10-24	0	Not recorded
2026-10-25	0	Not recorded
2026-10-26	0	Not recorded
2026-10-27	0	Not recorded
2026-10-28	0	Not recorded
2026-10-29	0	Not recorded
2026-10-30	0	Not recorded
2026-10-31	0	Not recorded
2026-11-01	0	Not recorded
2026-11-02	0	Not recorded
2026-11-03	0	Not recorded
2026-11-04	0	Not recorded
2026-11-05	0	Not recorded
2026-11-06	0	15

Figure 17: TC-BURN-01: today's actual value (15) after a 5-point task was completed, shown on the chart and in its table.

Appendix C: Re-running the Suite

```
npm ci
npx playwright install firefox     # once; Chrome and Edge use the installed browsers
npm run build
npm run preview                    # terminal 1: serves http://localhost:4173/
npm run test:system                # terminal 2: writes test-results/system/
```

The build must be made without Supabase credentials (no `.env` file), so that no test writes to a real database. `BROWSERS=chrome` runs Chrome only, and `ONLY=TC-KAN-02,TC-PERF-01` re-runs chosen cases.

Two behaviours of browser automation shaped the script; neither is a SprintFlow defect. Under Chrome automation a card that has already been dragged once does not start a second drag, so each timing sample drags a different card. Automation also cannot scroll during a drag, so drops land on the target column at the dragged card's height. Firefox has neither limitation.